Temperature practical

Plan a fair comparison

1 Choose the variable

Example hypothesis: raising acid concentration increases the temperature rise during neutralisation until alkali limits reaction.

Independent: acid concentration.
Dependent: temperature change.

Keep acid/alkali volumes and alkali concentration fixed.

2 Prepare & control

Polystyrene cup supported in beaker; lid and thermometer. Use fresh measured solutions at the same starting temperature.

Control apparatus, initial temperature and stirring/recording method. Goggles; handle chemicals as instructed.

Measure the change

3 Mix & record

Record starting temperature. Mix, replace lid, stir consistently; record at regular intervals.

Exothermic: find maximum. Endothermic: find minimum.

Change = extreme − initial. 20.5 → 28.0 °C gives **+7.5 °C.**

4 Repeat & plot

Repeat each condition, investigate anomalies and calculate a mean.

Plot concentration on x-axis, temperature change on y-axis.

A plateau may occur when alkali becomes limiting. More added acid cannot react after alkali is used up.

Interpret the evidence

5 Explain improvements

Insulation/lid reduce energy transfer to surroundings. Stirring evens out mixture temperature.

Prompt readings avoid missing peak. Repeats reduce random error effects, not thermometer calibration error.

If given time data, extrapolate appropriate trends to mixing time to estimate change before heat loss.

6 Apply the method

Other reactions: displacement; acid + metal; acid + carbonate. Decide appropriate variable and controls.

Allow gases to escape safely; keep flames away from hydrogen.

Compare applications using measured temperature change, duration, cost and safe operating temperature.